Model Parameterization and Literature Justification: Milkweed–Monarch–Bumblebee System

Project Parameterization Report

November 2026

1 Background and Prior Literature

To ground our differential equation model in empirical ecology, we adopt the common milkweed (*Asclepias syriaca*), monarch butterfly caterpillar (*Danaus plexippus*), and bumblebee (*Bombus* spp.) as our tri-trophic model system. Previous theoretical work by Revilla (2015) modeled plant–herbivore interactions under logistic plant growth and linear herbivory; our Stage 2 model extends this framework by incorporating saturating Type II pollinator visitation and density-dependent pollinator competition.

2 Dimensional Consistency Verification

We verify that all terms in the Stage 2 system maintain dimensional consistency in units of plants $\cdot$ day^{-1}, caterpillars $\cdot$ day^{-1}, and bees $\cdot$ day^{-1}:

$$\frac{dP}{dt} = \underbrace{rP\left(1-\frac{P}{K}\right)}_{\text{day}^{-1}\cdot\text{plants}} - \underbrace{aPC}_{(\text{cat}\cdot\text{day})^{-1}\cdot\text{plants}\cdot\text{cat}} + \underbrace{\beta\frac{vPB}{H+P}\left(1-\frac{P}{K}\right)}_{\frac{\text{plants}}{\text{visit}}\cdot\frac{\text{visits}}{\text{bee}\cdot\text{day}}\cdot\frac{\text{plants}\cdot\text{bees}}{\text{plants}}} \quad \left[\frac{\text{plants}}{\text{day}}\right] \tag{1}$$

$$\frac{dC}{dt} = \underbrace{\epsilon aPC}_{\frac{\text{cat}}{\text{plant}}\cdot(\text{cat}\cdot\text{day})^{-1}\cdot\text{plants}\cdot\text{cat}} - \underbrace{\mu C}_{\text{day}^{-1}\cdot\text{cat}} \quad \left[\frac{\text{caterpillars}}{\text{day}}\right] \tag{2}$$

$$\frac{dB}{dt} = \underbrace{\gamma\frac{vPB}{H+P}}_{\frac{\text{bees}}{\text{visit}}\cdot\frac{\text{visits}}{\text{bee}\cdot\text{day}}\cdot\frac{\text{plants}\cdot\text{bees}}{\text{plants}}} - \underbrace{dB}_{\text{day}^{-1}\cdot\text{bees}} - \underbrace{qB^2}_{(\text{bee}\cdot\text{day})^{-1}\cdot\text{bees}^2} \quad \left[\frac{\text{bees}}{\text{day}}\right] \tag{3}$$

3 Baseline Parameter Values

Table 1 summarizes the baseline parameter estimates derived from empirical studies on *Asclepias*, *Danaus plexippus*, and *Bombus* foraging dynamics.

Table 1: Baseline biological parameters for the Milkweed–Monarch–Bumblebee system.

Symbol	Value	Units	Biological Meaning & Literature Justification
r	0.015	day^{-1}	**Intrinsic plant growth rate:** Estimated via $r = \ln(2)/T_{\text{double}}$ assuming an uncrowded clonal milkweed patch doubles in ramet count over ~ 46 days during the growing season (range: 0.0011–0.015 day^{-1}; Xerces Society, 2018).
K	50	plants	**Plant carrying capacity:** Baseline patch size of 50 milkweed ramets.
a	0.00138	$(\text{cat} \cdot \text{day})^{-1}$	**Herbivory loss rate:** Fraction of plant biomass consumed per caterpillar per day, scaled from empirical leaf area (94.3 cm^2) and ramet consumption assays (Machado et al., 2023; Grant et al., 2022).
ϵ	0.40	$\text{cat} \cdot \text{plant}^{-1}$	**Caterpillar conversion efficiency:** Accounts for average larval survival ($\sim 58\%$ across nine *Asclepias* species; Pocius et al., 2018) and biomass conversion (< 1).
μ	0.069	day^{-1}	**Caterpillar exit/mortality rate:** Baseline larval stage duration of ~ 14.5 days ($\mu \approx 1/14.5$; Pocius et al., 2018).
v	2428	$\text{visits} \cdot (\text{bee} \cdot \text{day})^{-1}$	**Maximum bee visitation rate:** Based on 346.9 $\text{min} \cdot \text{day}^{-1}$ foraging time (5.9 trips/day $\times$ 58.5 min/trip) at ~ 7 $\text{visits} \cdot \text{min}^{-1}$ (Minahan & Brunet, 2018).
H	25	plants	**Half-saturation plant abundance:** Set to $0.5K$, the plant density at which bee visitation reaches half its maximum rate.
β	0.0005	$\text{plants} \cdot \text{visit}^{-1}$	**Plant growth per bee visit:** Derived from 0.0149 pods/flower (0.93 pods/62.5 flowers; Woods et al., 2012) multiplied by a 1–5% seedling establishment rate (0.00015–0.00075).
γ	2.96×10^{-5}	$\text{bees} \cdot \text{visit}^{-1}$	**Bee growth per flower visit:** Calibrated to balance natural mortality at carrying capacity ($\gamma = \frac{d}{v \cdot K/(H+K)}$), representing colony resource conversion into new workers.
d	0.048	day^{-1}	**Natural bee mortality rate:** Inverse of mean bumblebee worker lifespan of 20.95 days (range: 17.6–30 days; Da Silva-Matos & Garófalo, 2000).
q	0.0041	$(\text{bee} \cdot \text{day})^{-1}$	**Density-dependent bee competition:** Calibrated from $q = \left(\gamma \frac{vP}{H+P} - d\right)/B$ for a foraging density of $B = 10$ bees in a high-density patch ($P = 40$).

4 Connection to Equilibrium Thresholds

Using Bella's analytical coexistence condition,

$$\frac{dH}{\gamma v - d} < \frac{\mu}{\epsilon a} < K, \tag{4}$$

our baseline parameters yield an equilibrium plant threshold of $P^* = \frac{\mu}{\epsilon a} = \frac{0.069}{(0.40)(0.00138)} = 125$ plants. This demonstrates that in a small patch ($K = 50 < 125$), monarch caterpillars cannot sustain a positive equilibrium (E_{PB} is stable instead), whereas in a larger meadow ($K > 125$), three-species coexistence (E_{PCB}) becomes biologically feasible.